Content:

Quantum and Nanoscale Physics of materials:

Physical interpretation of wave function, derivation of 1D time-independent Schrodinger equation (TISE) from classical wave equation, Quantum confinement effects - particle in a box, energy levels, Quantum tunnelling, Optical properties due to quantum confinement - blue shift, absorption, fluorescence. Numerical Problems.

Nanomaterials and its classification, Synthesis - Bottom-up approach - Solgel, hydrothermal. Top-down approach - Ball milling, lithography, laser ablation. Nano scale semiconductors—quantum dots, size-dependent optical and electrical characteristics. Applications in electronics, biomedicine, and solar cell technologies. Numerical Problems.

Introduction:

At the beginning of the 20th century, Newton's laws of motion were able to successfully describe the motion of the particles in classical mechanics (the world of large, heavy and slow bodies) and Maxwell's equations explained phenomena in classical electromagnetism. However the classical theory does not hold in the region of atomic dimensions. It could not explain the stability of atoms, energy distribution in the black body radiation spectrum, origin of discrete spectra of atoms, etc. It also fails to explain the large number of observed phenomena like photoelectric effect, Compton Effect, Raman Effect, Quantum Hall effect, superconductivity etc. **The insufficiency of classical mechanics led to the development of quantum mechanics (QM).**

Quantum mechanics gives the description of motion and interaction of particles in the small scale atomic system where the

discrete nature of the physical world becomes important. With the application of quantum mechanics, most of the outstanding problems have been solved. The direct implications of QM though very subtle have dramatic effect in our day to day life. For example, QM successfully introduced the concept of discrete energy which led to the conception and advancements in quantum computing, it also paved way for efficient long distance transfer of large quantities of data electronically. By understanding the electronic spin and related atomic properties, QM has aided in realizing energy efficient materials that can be applied to consumable electronic equipment and in commercial transport vehicles like Hyper loop, Maglev trains among others.

Black Body Radiation

Black-body radiation is the type of electromagnetic **radiation** emitted by a **black body** (an opaque and non-reflective **body**) held at constant, uniform temperature. In nature there are no perfect black bodies.

Black Body Spectrum: It is a graph showing the variation of the energy of the black body radiations as a function of their wavelengths or frequencies.The energy distribution in the black body spectrum is explained by **Wien's distribution law** in the lower wavelength region and **Rayleigh Jean's law** explains the energy distribution in the larger wavelength region.

Wien's law:

$$E_{\lambda} d\lambda = A\lambda^{-5} e^{\frac{-B}{\lambda T}} d\lambda$$

Rayleigh Jeans law:

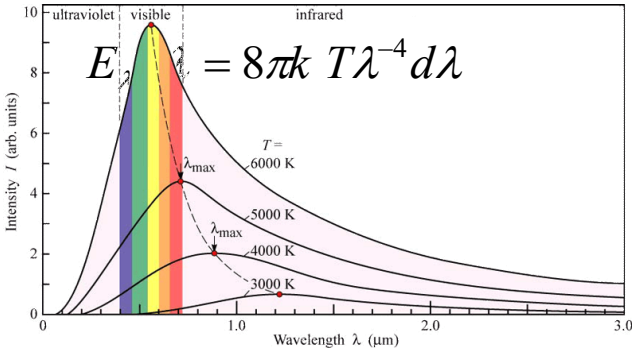


Fig: Blackbody Radiation spectrum

Neither Wien's law nor Rayleigh- jean's law could explain the energy distribution in the entire blackbody spectrum. The energy distribution in the entire blackbody spectrum was successfully explained by Max. Planck's quantum Theory.

Planck's quantum theory

The energy distribution in the black body radiation spectrum was successfully explained by Max Planck in the year 1900. According to Planck's quantum theory thermal energy is not emitted or absorbed continuously, but it is emitted or absorbed in discrete quantities called quanta. Each quanta has an energy 'hν' where h is the Planck's constant. Applying the Planck's quantum theory an expression for the energy distribution in the black body spectrum was obtained and it is called Planck's formula.

The Planck's formula is as follows

$$E_\lambda d\lambda = \frac{8\pi hc}{\lambda^5} \frac{1}{e^{(hc/\lambda kT)} - 1} d\lambda$$

Where k is the Boltzmann's constant; h- Planck's constant and c is the velocity of light, λ is the wavelength of the black-body radiation and ω is the angular frequency of light.

Photoelectric effect:

When the light of a **suitable wavelength** shines on certain materials, then electrons are **spontaneously** emitted from the surface of material. It can be observed in any material but most readily in metals and good conductors. This phenomenon is known as **the photoelectric effect**.

The materials that exhibit photoelectric effect are called photosensitive materials and the emitted electrons are called photoelectrons. Heinrich Hertz first observed this phenomenon in 1887.

The electrons are emitted only when the photons reach or exceed a threshold frequency (energy) and below that threshold, no electrons are emitted from the metal regardless of the light intensity or the length of time of exposure to the light. To explain this phenomenon, Albert Einstein proposed that light be seen as a collection of discrete bundles of energy (photons), each with energy $h\upsilon$, where υ is the frequency of the light that is being quantized and h is known as the Planck constant.

Einstein's photoelectric equation:

Einstein, in 1905, proposed that the light energy is localized in small packets similar to the Planck's idea of quanta, and named such packets as photons. According to Einstein, in photoelectric effect one photon is completely absorbed by one electron, which thereby gains the quantum of energy and may be emitted from the metal.

Thus, the photon energy is used in the following two parts:

i). A part of its energy is used to free the electron from the atoms of the metal surface. This energy is known as a photoelectric work function of metal (W_o)

ii) The other part is used in giving kinetic energy (½ mv^2) to the electron.

Thus $h\nu = W_o + \frac{1}{2}mv^2$ Where 'v' - velocity of the emitted electron.

This equation is known as Einstein's photoelectric equation.

When the photon's energy is of such a value that it can just liberate the electron from metal, then the kinetic energy of the electron will be zero. Then the above equation reduces to $h\nu_o = W_o$, where ν_o is called the threshold frequency.

Threshold frequency is defined as the minimum frequency which can cause photoelectric emission. Below this frequency no emission of electron takes place.

Compton Effect:

When a monochromatic beam of high frequency radiation (X – rays, γ – rays, etc.) is scattered by a substance, then the scattered radiation contains two components - one having a lower frequency or greater wavelength called as modified radiation and the other having the same frequency or wavelength called as unmodified radiation. This phenomenon is known as Compton Effect and was discovered by Prof. A.H. Compton in 1921. The process of recoiling of electron and scattering of photon is as shown in the following figure:

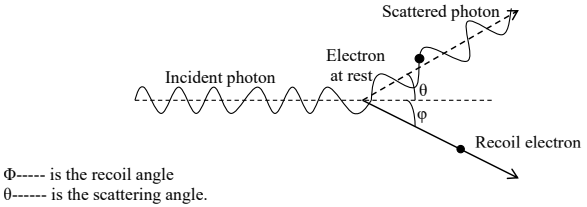


Fig: Schematic diagram of Compton Effect

According to the quantum concept of radiation, the radiation is constituted by energy packets called photons. The energy of photon is hν, where h is Planck's constant and ν is the frequency of radiation. The photons move with velocity of light c, possess momentum hν/c and obey all the laws of conservation of energy and momentum. According to Compton, the phenomenon of scattering is due to an elastic collision between two particles, the photon of incident radiation and the electron of the scatterer. When the photon of energy hν collides with the electron of the scatterer at rest, it transfers some energy to the electron, i.e., it loses the energy. The scattered photon will therefore have a smaller energy and consequently a lower frequency (ν') or greater wavelength(λ') than that of the incident photon. The observed change in frequency or wavelength of the scattered radiation is known as Compton Effect. In the scattering process, the electron gains kinetic energy and thus recoils with a velocity v. The changed wavelength of the photon scattered through an angle θ is given by

$$\Delta\lambda = (\lambda' - \lambda) = \frac{h}{m_0 c}[1 - \cos\vartheta]$$

Wave Packet

A **wave packet** refers to the case where two (or more) waves exist simultaneously. A wave packet is often referred to a wave group. This situation is permitted by the principle of superposition. In physics, a **wave packet** (or **wave train**) is a short "burst" or "envelope" of localized wave action that travels as a unit. A wave packet can be analyzed into, or can be synthesized from, an infinite set of component sinusoidal waves of different wavenumbers, with phases and amplitudes such that they interfere constructively only over a small region of space, and destructively elsewhere. Each component wave function, and hence the wave packet, are solutions of a wave equation. Depending on the wave equation, the wave packet's profile may remain constant (no dispersion) or it may change (dispersion) while propagating.

Phase velocity and Group velocity

A monochromatic wave has velocity called the **phase velocity**

$$V_p = \frac{\omega}{k} = \upsilon\lambda$$

where ω is the angular frequency, $k = \frac{2\pi}{\lambda}$ is the wave number, υ is the frequency.

However, if we have a compound wave (wave packet) that is composed of individual waves with a range of frequencies, each individual wave has its phase velocity, but the amplitudes of the waves add up to produce a wave packet which has a velocity all its own. This velocity is called the **group velocity** (v_g) and is usually different from the individual phase velocities of the waves that make up the wave packet. Schrodinger postulated that moving particle (electron, proton..etc) is equivalent to a wavepacket rather than a single wave. The velocity with which the resultant envelope of the

group waves travels is called group velocity (v_g) and is equal to the particle velocity (v).

It is given by the expression

$$V_g = \frac{d\omega}{dk} = V$$

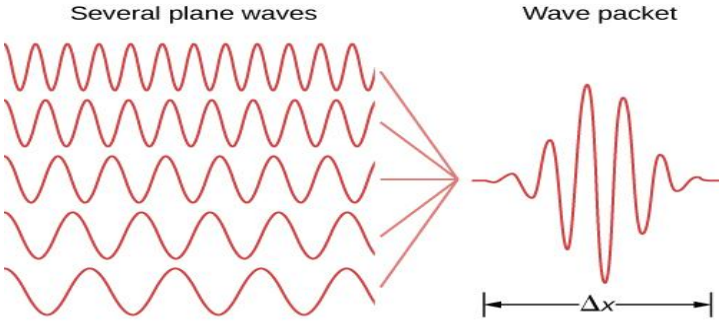


Fig: Schematic diagram of Wave Packet

The equation that describes the evolution of this wave packet is the Schrödinger equation. It can be Time independent or Time dependent.

Wave and particle duality of radiation

To understand the wave and particle duality, it is necessary to know what a particle is and what a wave is.

A particle is a localized mass and it is specified by its mass, velocity, momentum, energy, etc. In contrast a wave is a spread out disturbance. A wave is characterized by its wavelength, frequency, velocity, amplitude, intensity, etc. It is hard to think mass being associated with a wave. Considering the above facts, it appears difficult to accept the conflicting ideas that radiation has a wave particle duality. However this acceptance is essential because the radiation exhibits phenomena like interference, diffraction, polarization, etc., and shows the wave nature and it also exhibits the

particle nature in the black-body radiation effect, photoelectric effect, Compton Effect etc.

Radiation, thus, sometimes behave as a *wave* and at some other time as a *particle*, this is the wave particle duality of radiation.

De Broglie's concept of matter waves.

Louis de-Broglie in 1924 extended the wave particle dualism of radiation to fundamental entities, such as electrons, protons, neutrons, atoms, molecules, etc. de-Broglie put a bold suggestion that like radiation, matter also has dual characteristic, at a time when there was absolutely no experimental evidence for wave like properties of matter waves. **de-Broglie Hypothesis** of matter waves is as follows.

- In nature energy manifests itself in two forms, namely matter and radiation.
- Nature loves symmetry.
- As radiation can act like both wave and a particle, material particles (like electrons, protons, etc.) in motion should exhibit the property of waves.

These waves due to moving matter are called *matter waves or de-Broglie waves or pilot waves.*

Wavelength of matter waves:

The concept of matter waves is well understood by combining Planck's quantum theory and Einstein's theory. Consider a photon of energy E, frequency γ and wavelength λ.

By Planck's theory $E = h\gamma = \frac{hc}{\lambda}$

By Einstein's mass-energy relation $E = mc^2$

By equating and rearranging the above equations, we get

$$mc^2 = \frac{hc}{\lambda}$$

$$\lambda = \frac{h}{mc} = \frac{h}{p}$$

Where, p is the momentum of the photon and h is a Planck's constant.

Now consider a particle of mass m moving with a velocity v and momentum p. According to the de-Broglie hypothesis matter also has a dual nature. Hence the wavelength λ of matter waves is given by

$$\lambda = \frac{h}{mv} = \frac{h}{p}$$

This is the equation for the de-Broglie wavelength.

de Broglie wavelength of an electron

Consider an electron of mass 'm' accelerated from rest by an electric potential V. The electrical work done (eV) is equal to the kinetic energy E gained by the electron.

$$E = eV$$

$$\therefore E = \frac{1}{2}mv^2$$

$$\therefore m^2v^2 = 2mE$$

$$mv = \sqrt{2mE}$$

Therefore wavelength of electron wave

$$\lambda = \frac{h}{p} = \frac{h}{mv}$$

$$\therefore \lambda = \frac{h}{\sqrt{2mE}}$$

But $E = eV$

$$\therefore \lambda = \frac{h}{\sqrt{2meV}} = \frac{h/\sqrt{2me}}{\sqrt{V}}$$

Substituting the values of e, m and h, we get

$$\frac{h}{\sqrt{2me}} = 12.28 A^o$$

$$\therefore \lambda = \frac{12.28}{\sqrt{V}} A^o$$

Note: Instead of a an electron, if a particle of charge 'q' is accelerated through a potential difference V, then

$$\lambda = \frac{h}{\sqrt{2mqV}}$$

Heisenberg's uncertainty principle

Heisenberg's uncertainty principle is a direct consequence of the dual nature of matter. In classical mechanics, a moving particle at any instant has a fixed position in space and a definite momentum which can be determined if the initial values are known **(we can know the future if we know the present)**

In wave mechanics a moving particle is described in terms of a wave group or wave packet. According to Max Born's probability interpretation the particle may be present anywhere inside the wave packet. When the wave packet is large, the momentum can be fixed, but there is a large uncertainty in its position. On the other hand, if the wave packet is small the position of the particle may be fixed, but the particle will spread rapidly and hence the momentum (or velocity) becomes indeterminate. In this way certainty in momentum involves uncertainty in position and the certainty in

position involves uncertainty in momentum. Hence it is impossible to know within the wave packet where the particle is and what is its exact momentum. **(We cannot know the future because we cannot know the present)**. Thus we have Heisenberg's uncertainty principle.

According to the Heisenberg's uncertainty principle *"It is impossible to specify precisely and simultaneously certain pairs of physical quantities like position and momentum that describe the behavior of an atomic system"*. Qualitatively, this principle states that in any simultaneous measurement the product of the magnitudes of the uncertainties of the pairs of physical quantities is equal to or greater than h/4π (or of the order of h)

Considering the pair of physical quantities such as position and momentum, we have

$$\Delta p\, \Delta x \geq h/4\pi \qquad \ldots\ldots 1$$

Where Δp and Δx are the uncertainties in determining the momentum and the position of the particle.

Similarly, we have other canonical forms as

$$\Delta E \Delta t \geq h/4\pi \qquad \ldots\ldots 2$$

$$\Delta J \Delta \theta \geq h/4\pi \qquad \ldots\ldots 3$$

Where ΔE and Δt are uncertainties in determining energy and time while ΔJ and $\Delta\theta$ are uncertainties in determining angular momentum and angular position.

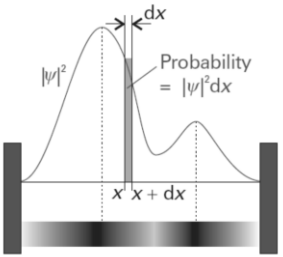
Physical significance of the wave function

The wave function ψ(x, t) is the solution of Schrödinger wave equation. It gives a quantum-mechanically complete description of the behavior of a moving particle. The wave function ψ cannot be measured directly by any physical experiment. However, for a given ψ, knowledge of usual dynamic variables, such as position, momentum, angular momentum, etc., of the particle is obtained by performing suitable mathematical operations on it.

The most important property of ψ is that it gives a measure of the probability of finding a particle at a particular position. ψ is also called the **probability amplitude**. In general ψ is a complex quantity, whereas the probability must be real and positive. Therefore a term called probability density is defined. The **probability density P (x,t)** is a product of the wave function ψ and its complex conjugate ψ^*.

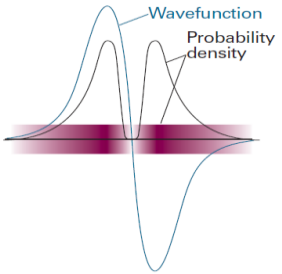
$$\therefore P(x,t) = \psi\psi^* = |\psi(x,t)|^2$$

The Born interpretation of the wavefunction



The interpretation of the wave function in terms of the location of the particle is based on a suggestion made by Max Born. He made use of an analogy with the wave theory of light, in which the square of the amplitude of an electromagnetic wave in a region is

interpreted as its intensity and therefore (in quantum terms) as a measure of the probability of finding a photon present in the region. The Born interpretation of the wavefunction focuses on the square of the wavefunction (or the square modulus, $|\psi|^2 = \psi^*\psi$, if ψ is complex). It states that the value of $|\psi|^2$ at a point is proportional to the probability of finding the particle in a region around that point. If the wavefunction of a particle has the value ψ at some point x, then the probability of finding the particle between x and x + dx is proportional to $|\psi|^2 dx$. Thus, $|\psi|^2$ is the probability density, and to obtain the probability it must be multiplied by the length of the infinitesimal region dx. The wavefunction ψ itself is called the probability amplitude. The Born interpretation does away with any worry about the significance of a negative (and, in general, complex) value of ψ because $|\psi|^2$ is real and never negative. There is no direct significance in the negative (or complex) value of a wavefunction: only the square modulus, a positive quantity, is directly physically significant, and both negative and positive regions of a wavefunction may correspond to a high probability of finding a particle in a region.



Normalization of wave function

If ψ is a wave function associated with a particle, then $\psi^2 d\tau$ is the probability of finding the particle in a small volume $d\tau$. If it is certain that the particle is present in a volume τ then the total

probability in the volume τ is unity i.e., $\int_{\tau} \psi^2 d\tau = 1$. This is the normalization condition.

In one dimension the normalization condition is $\int_x \psi^2 dx = 1$

Note: When the particle is bound to a limited region the probability of finding the particle at infinity is zero i.e., $\psi\psi^*$ at $x = \infty$ is zero.

Properties of wave function:

The wave function ψ should satisfy the following properties to describe the characteristics of matter waves.

1. ψ must be a solution of Schrödinger wave equation.
2. The wave function ψ should be continuous and single valued everywhere. Because it is related to the probability of finding a particle at a given position at a given time, which will have only one value.
3. The first derivative of ψ with respect to x should be continuous and single valued everywhere, since it is related to the momentum of the particle which should be finite.
4. Ψ must be normalized so that ψ must go to 0 as $x \to \pm\infty$, so that $\int \psi^2 d\tau$ over all the space be a finite constant.

SCHRÖDINGER'S WAVE EQUATION

In1926 Schrödinger starting with de-Broglie equation (λ = h/mv) developed it into an important mathematical theory called wave mechanics which found a remarkable success in explaining the behavior of the atomic system and their interaction with electromagnetic radiation and other particles. In water waves, the quantity that varies periodically is the height of water surface. In sound waves it is pressure. In light waves, electric and magnetic

fields vary. *The quantity whose variation gives matter waves is called* ***wave function (ψ).***

The value of wave function associated with a moving body at a particular point x in space at a time t is related to the likelihood of finding the body there at a time. A wave function ψ(x,t) that describes a particle with certain uncertainty in position, moving in a positive x-direction with precisely known momentum and kinetic energy may assume any one of the following forms: Sin(ωt -kx) , cos(ωt -kx), $e^{i(\omega t - kx)}$, $e^{-i(\omega t - kx)}$or some linear combinations of them.

Schrödinger wave equation is the wave equation of which the wave functions are the solutions. **It cannot be derived from any basic principles,** but can be arrived at, by using the de-Broglie hypothesis in conjunction with the classical wave equation.

Time Independent one dimensional Schrödinger wave equation (TISE)

In many situations the potential energy of the particle does not depend on time explicitly; the force that acts on it, and hence the potential energy vary with the position of the particle only. The Schrödinger wave equation for such a particle is time independent wave equation. Let ψ(x,t) be the wave function of the matter wave associate with a particle of mass m moving with a velocity v. The differential equation of the wave motion is as follows.

$$\frac{\partial^2 \psi}{\partial x^2} = \frac{1}{v^2}\frac{\partial^2 \psi}{\partial t^2} \text{.....(1)}$$

The solution of the Eq.(1) as a periodic displacement of time t is

$$\psi(x,t) = \psi_0(x)e^{-i\omega t} \text{ .....(2)}$$

Where $\psi_0(x)$ is the amplitude of the matter wave.

Differentiating Eq.2 partially twice w.r.t. to t, we get

$$\frac{\partial \psi}{\partial t} = -i\omega\, \psi_0(x)\, e^{-i\omega t}$$

$$\frac{\partial^2 \psi}{\partial t^2} = i^2\omega^2\, \psi_0(x)\, e^{-i\omega t}$$

$$\frac{\partial^2 \psi}{\partial t^2} = -\omega^2\, \psi_0(x) e^{-i\omega t} \quad \ldots\ldots. (3)$$

Performing partial derivative of (2) w.r.t x, we get

$$\frac{\partial^2 \psi}{\partial x^2} = \frac{\partial^2 \psi_0}{\partial x^2}\, e^{-i\omega t} \quad \ldots\ldots\ldots\ (4)$$

Substituting Eq.3 and 4 in Eq.1 we get

$$\frac{\partial^2 \psi_0}{\partial x^2}\, e^{-i\omega t} = -\frac{\omega^2}{v^2}\psi_0 e^{-i\omega t}$$

$$\frac{\partial^2 \psi_0}{\partial x^2} = -\frac{\omega^2}{v^2}\psi_0 \quad \ldots\ldots.. (5)$$

We will drop the ψ_0 notation in favour of ψ for simplicity from now on. We can further substitute the partial derivative by total derivative to get

$$\frac{d^2 \psi}{dx^2} = -\frac{\omega^2}{v^2}\psi \quad \ldots\ldots. (6)$$

We have $\frac{\omega^2}{v^2} = k^2 = \left(\frac{2\pi}{\lambda}\right)^2 = \frac{4\pi^2}{\lambda^2}$

Substituting this in Eq6, we get

$$\frac{d^2 \psi}{dx^2} = -\frac{4\pi^2}{\lambda^2}\psi \quad \ldots\ldots(7)$$

$$\frac{d^2\psi}{dx^2} + \frac{4\pi^2}{\lambda^2}\psi = 0 \quad \ldots\ldots(8)$$

Substituting the wavelength of the matter waves λ=h/mv in Eq.6 we get

$$\frac{d^2\psi}{dx^2} + 4\pi^2\frac{m^2v^2}{h^2}\psi = 0 \quad \ldots\ldots\ldots(9)$$

If E and V are the total and potential energies of the particle respectively, then the kinetic energy of the particle

$$E = \frac{1}{2}mv^2 = E - V$$

$$\therefore m^2v^2 = 2m(E - V)$$

Substituting this in Eq.9, we get

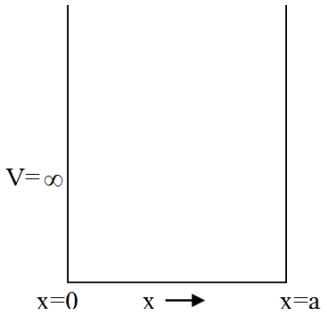
$$\frac{d^2\psi}{dx^2} + \frac{8\pi^2 m}{h^2}(E - V)\psi = 0 \quad \ldots\ldots\ldots (10)$$

Hence ψ is a function of *x* alone and is independent of time. This equation is called the Schrödinger time- independent one dimensional wave equation.

QUANTUM CONFINEMENT

1. For a Particle in a one-dimensional potential well of infinite depth (Particle in a box)

Consider a particle of mass 'm' moving freely in x- direction in the region from x=0 to x=a. Outside this region potential energy 'V' is infinity and within this region V=0.Outside the box Schrodinger's wave equation is



$$\frac{d^2\Psi}{dx^2} + \frac{8\pi^2 m}{h^2}(E-\infty)\Psi = 0 \quad \text{------ (1)}$$

This equation holds good only if ψ =0 for all points outside the box i.e., $|\Psi|^2 = 0$, which means that the particle cannot be found at all outside the box. Inside the box V = 0, hence the Schrodinger's equation is given by,

$$\frac{d^2\Psi}{dx^2} + \frac{8\pi^2 mE}{h^2}\Psi = 0$$

$$\frac{d^2\Psi}{dx^2} + k^2\Psi = 0 \quad \text{-------- (2)}$$

$$k^2 = \frac{8\pi^2 m}{h^2} \quad \text{----- (3)}$$

Discussion of the solution

The solution of the above equation 2 is given by

$$\Psi = A\cos kx + B\sin kx \text{.....(4)}$$

Where A & B are constants which depending on the boundary conditions of the well.

Now apply boundary conditions for this,

Condition: I at x =0, ψ = 0.

Substituting the condition I in the equation 4, we get A =0 and B ≠ 0. (If B is also zero for all values of x, ψ is zero. This means that the particle is not present in the well.)

Now the equation 4 can be written as $\Psi = B\sin kx$.....(5)

Condition: II at x =a, ψ = 0

Substituting the condition II in equation 5 we get

0= B sin(ka)

Since $B \neq 0$,

$\sin ka = 0 \qquad \therefore ka = n\pi$

$$k = \frac{n\pi}{a} \qquad k^2 = \frac{n^2\pi^2}{a^2} \quad where,\ n = 1,2,3...$$

Substitute the value of k^2 in equation (3).

$$\frac{8m\pi^2 E}{h^2} = \frac{n^2\pi^2}{a^2}$$

$$E = \frac{n^2h^2}{8ma^2} .....(6)$$

The equation (6) gives energy values or Eigen value of the particle in the well. When n=0, $\psi_n = 0$. This means to say that the electron is not present inside the box, which is not true. Hence the lowest value of 'n' is 1.

∴ The lowest energy corresponds to 'n'=1 is called the zero-point energy or Ground state energy.

$$E_{zero-point} = \frac{h^2}{8ma^2}$$

All the states of n>1 are called excited states.

To evaluate B in equation (3), one has to perform normalization of wave function.

Normalization of wave function:

Consider the equation, $\psi = B\sin kx = \mathrm{B}\sin\frac{\mathrm{n}\pi}{\mathrm{a}}x$

The integral of the wave function over the entire space in the box must be equal to unity because there is only one particle within the box, the probability of finding the particle is 1.

$$\int_0^a |\Psi|^2 dx = 1$$

$$B^2\int_0^a \sin^2\frac{n\pi}{a}xdx = 1$$

But $\sin^2\theta = \frac{1}{2}(1-\cos 2\theta)$

$$\frac{B^2}{2}\int_0^a\left(1-\cos\frac{2n\pi x}{a}\right)dx=1`$$

$$\frac{B^2}{2}\left[\int_0^a dx-\int_0^a\cos\frac{2n\pi x}{a}dx\right]=1$$

$$\frac{B^2}{2}\left[x-\frac{a}{2n\pi}\sin\left(\frac{2n\pi x}{a}\right)\right]_0^a=1$$

$$\frac{B^2}{2}\left[a-\frac{a}{2n\pi}\sin 2n\pi-0\right]=1$$

$$\frac{B^2a}{2}=1 \qquad \therefore B=\sqrt{2/a}$$

Thus the normalized wave function of a particle in a one-dimensional box is given by,

$$\Psi_n=\sqrt{\frac{2}{a}}\sin\left(\frac{n\pi}{a}\right)x \quad \text{where, n=1,2,3……}$$

This equation gives the Eigen functions of the particle in the box. The Eigen functions for n=1,2,3..are as follows.

$$\Psi_1=\sqrt{\frac{2}{a}}\sin\left(\frac{\pi}{a}\right)x$$

$$\Psi_2=\sqrt{\frac{2}{a}}\sin\left(\frac{2\pi}{a}\right)x$$

$$\Psi_3=\sqrt{\frac{2}{a}}\sin\left(\frac{3\pi}{a}\right)x$$

Since the particle in a box is a quantum mechanical problem we need to evaluate the most probable location of the particle in a box and its energies at different permitted state.

Wave functions and Probabilities

Let us consider the three lowest energy solutions:

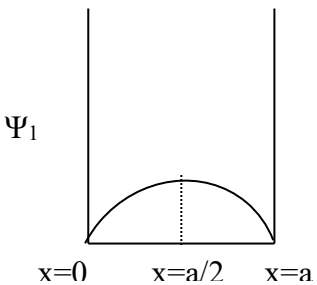
Case (1): n=1

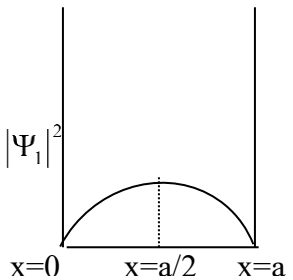
This is the ground state and the particle is normally found in this state. For n=1, the Eigen function is

$$\Psi_1 = \sqrt{\frac{2}{a}}\sin\left(\frac{\pi}{a}\right)x$$

In the above equation Ψ =0 for both x=0 & x=a. But Ψ_1 has a maximum value for x=a/2.

$$\Psi_1 = \sqrt{\frac{2}{a}}\sin\left(\frac{\pi}{a}\right)\frac{a}{2} = \sqrt{\frac{2}{a}} \quad \text{and} \quad \psi_1^2 = \frac{2}{a}$$





A plot of $|\Psi_1|^2$ the probability density versus 'x' is as shown. From the figure, it indicates the probability of finding the particle at different locations inside the box.

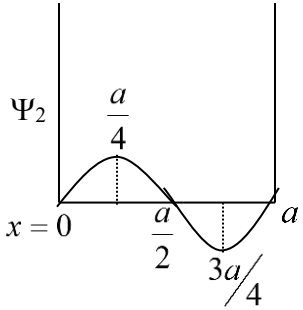
$\therefore |\Psi_1|^2$=0 at x = 0 and x = a, also $|\Psi_1|^2$ is maximum at x = (a/2).

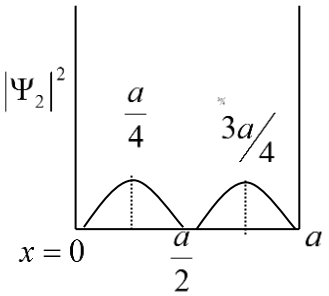
This means that in the ground state the particle cannot be found at the walls of the box and the probability of finding the particle is maximum at the central region. The Energy in the ground state is given by $E_1 = \frac{h^2}{8ma^2}$.

Case (2): n=2

This is the first excited state. The Eigen function for this state is given by $\Psi_2 = \sqrt{\frac{2}{a}} \sin\left(\frac{2\pi}{a}\right)x$

Now, $\Psi_2 = 0$ for the values $x = 0, a/2, a$ and Ψ_2 reaches maximum for the values $x = a/4, 3a/4$. These facts are seen in the following plot.





From the figure it can be seen that

$|\Psi_2|^2 = 0$ at x = 0, $a/2, a$ and $|\Psi_2|^2 = 2/a$ at $x = a/4, 3a/4$

This means that in the first excited state the particle cannot be observed either at the walls or at the center. The energy is $E_2 = 4E_1$. Thus the energy in the first excited state is 4 times the zero point energy.

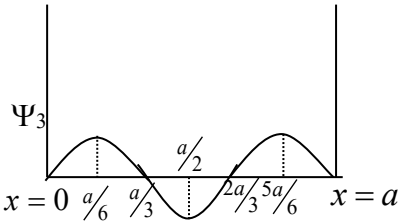
Case 3: n =3.

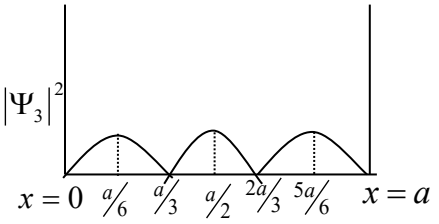
This is the second excited state and the Eigen function for this state is given by

$$\Psi_3 = \sqrt{\frac{2}{a}} \sin\left(\frac{3\pi}{a}\right)x$$

Now, Ψ_3 =0 for the values $x = 0, a/3, 2a/3, a$

and Ψ_3 reaches maximum $\sqrt{\frac{2}{a}}$ at $x = a/6, a/2, 5a/6$





$|\Psi_3|^2$ =0 for the values $x = 0, a/3, 2a/3, a$ and $|\Psi_3|^2$ reaches maximum 2/a at $x = a/6, a/2, 5a/6$ at which the particle is most likely to be found. The energy corresponds to second excited state is given by $E_3 = 9E_1$.

2. Quantum Tunneling in 1 dimension (Finite Square Well)

In infinite square well potential, we know that the electrons never leave the well. This means that the probability of finding the particle at x= 0 and x = a and outside the well has to be zero.

So, V(0) = V(a) = ∞ and ψ(0) = ψ(a) = 0

A more realistic description of the confining potential is when depth of the well is finite. This ALLOWS the particle to be outside the well/box also, with finite probability. This phenomena is called Quantum Tunneling.

We now study the finite square well, shown in figure below.

$V(x) = -V_o$ for $|x| < a/2$, $V_0 > 0$, $V(x) = 0$ for $|x| < a/2$

The width of the well is a. We have placed the bottom of the well differently than in the case of the infinite square well. The bottom of the infinite square well was at zero potential energy. If we wanted to obtain the infinite square well as a limit of the finite square well we would have to take V_0 to infinity, but care is needed to compare energies. The ones in the infinite square well are measured with respect to a bottom at zero energy. The ones in the finite square well are measure with respect to a bottom at $-V_0$.

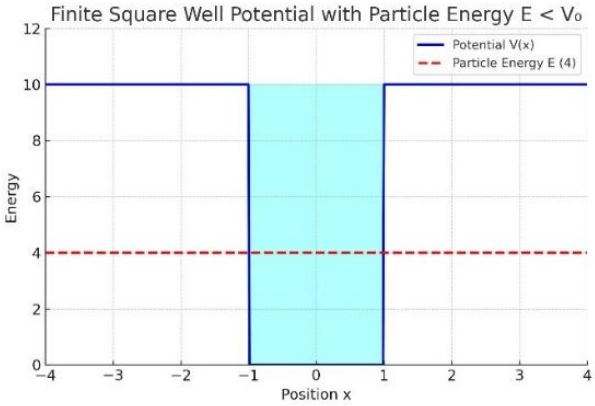
We will be interested in bound states namely, energy eigenstates that are normalizable. For this the energy E of the states must be negative. This is readily understood. If $E > 0$, any solutions in the region $|x| > a/2$, where the potential vanishes would be a plane wave, extending all the way to infinity. Such a solution would not be normalizable. The energy E is shown as a dashed line in the figure. We have $-V_0 < E < 0$. Note that since E is negative we have $E = -|E|$. For a bound state of energy E, the energy E measured with respect to the bottom of the potential is $E' = E - (-V_0) = V_0 - |E| > 0$. Those E' are the ones that can be compared with the energies of the infinite square well in the limit as $V_0 \rightarrow \infty$. What are the bound state solutions to the Schrodinger equation with this potential? We have to examine how the equation looks in the various regions where the potential is constant and then use boundary conditions to match the solutions across the points where the potential is discontinuous. Let's examine the regions, where, for simplicity, we define $\alpha(x)$ by

$$\frac{d^2\Psi}{dx^2} = -\frac{8\pi^2 m}{h^2}(E - V(x))\Psi(x) = \alpha(x)\Psi(x) \quad \text{.....(1)}$$

where we have defined the factor $\alpha(x)$ that multiplies the wavefunction on the right-hand side of the Schrodinger equation. We then consider the two regions • region $x > a$: $\alpha(x)$ is a positive constant.

The wavefunction in this region constructed with real exponentials. The region x < a: α(x) is a negative constant. The wavefunction in this region is constructed with trigonometric functions.

The potential V (x) for the finite square well is an even function of x: V (−x) = V (x). We use a theorem (without proof) that for an even potential the bound states are either odd or even functions (ψ(−x) = ψ(x)) of x. The symmetric finite well extends from x = -0.5 a to 0.5 a, as shown below, where a = 2, E = 4 and V_o= 10 in some arbitrary unit of position(x) and energy.



REGION I	x < -0.5a	$V(x) = V_o$
REGION II	-0.5a < x < 0.5a	V(x) = 0
REGION III	x > 0.5a	$V(x) = V_o$

Rewriting Equation 1 we get:

$$\frac{d^2\Psi}{dx^2} = \frac{8\pi^2 m}{h^2}(V(x) - E)\Psi \qquad \text{.....(2)}$$

In region II, $V(x)-E < 0$, hence

$$\frac{d^2\Psi}{dx^2} = -k^2\Psi \quad \text{------- (3)}$$

$$k^2 = \frac{8\pi^2 m}{h^2}(E - V(x))$$

As $V(x) = 0$ in region II, $k^2 = 8\pi^2 mE/h^2$, where k is a real number.

$$\psi_{II}(x) = A\cos(kx) \text{ OR } B\sin(kx) \quad \text{….(4)}$$

In regions I and III, $E < V(x)$ (This is the classically forbidden region for the particle(that is the classical particle cannot be found there) as kinetic energy is negative, which is not possible in classical mechanics. However this is possible in Quantum mechanics. The TISE in regions I and III become:

$$\frac{d^2\Psi}{dx^2} = \alpha^2(x)\Psi(x) \quad \text{……(5),}$$

where, $\alpha^2 = \frac{8\pi^2 m}{h^2}(V_o - E)$ ..... (6)

The solution in these two regions is given by :

$$\Psi_{III}(x) = Ce^{\alpha x} + De^{-\alpha x} \quad \text{…(7)}$$

$$\text{and } \Psi_I(x) = Ee^{\alpha x} + Fe^{-\alpha x} \quad \text{....(8)}$$

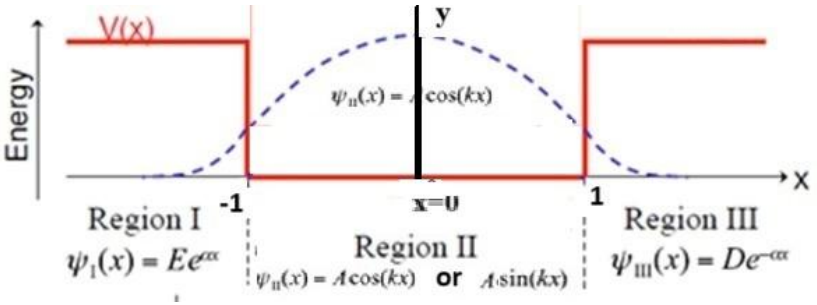
As $x \to \infty$, $e^{\alpha x}$ diverges, which is not possible since wave function has to be normalizable. Hence the divergent part is set to zero by putting $C = 0$. Thus we finally get

$$\Psi_{III}(x) = De^{-\alpha x} \quad \text{…..(9)}$$

As $x \to -\infty$, $e^{-\alpha x}$ diverges, which can be avoided by taking $F = 0$. Thus

$$\Psi_I(x) = Ee^{\alpha x} \quad \text{…..(10)}$$

The graphical illustration of these wave functions region wise is shown in the diagram below:



$\Psi_I(x) = Ee^{\alpha x}$, $\psi_{II}(x) = A\cos(kx)$ or $B\sin(kx)$, $\Psi_{III}(x) = De^{-\alpha x}$

If we want Even parity solution, as is applicable for the ground state, second, fourth, …excited states, we will use $\psi_{II}(x) = A\cos(kx)$.

However, if we want Odd parity solutions, as is applicable for the 1st, 3rd, 5th, ….excited states, we consider $\psi_{II}(x) = A\sin(kx)$.

Now, we have to match the wave-functions as well as their derivatives at the boundaries of these different regions.

$\Psi_I(-a/2) = \Psi_{II}(-a/2)$ and $(d\Psi_I/dx)_{x=-a/2} = (d\Psi_{II}/dx)_{x=-a/2}$ (11)

$\Psi_{II}(a/2) = \Psi_{III}(a/2)$ and $(d\Psi_{II}/dx)_{x=a/2} = (d\Psi_{III}/dx)_{x=a/2}$ (12)

Imposing continuity of wavefunction on cosine solution, we get:

from $\Psi_I(-a/2) = \Psi_{II}(-a/2)$: $Ee^{-\alpha a/2} = A\cos(ka/2)$ …… (13)

from $\Psi_{II}(a/2) = \Psi_{III}(a/2)$: $A\cos(ka/2) = De^{-\alpha a/2}$ ..... (14)

Dividing equation (13) by Equation (14)

we have $E = D$ (15)

Apply continuity of the first derivative, on the cosine solution.

From $(d\Psi_I/dx)_{x=-a/2} = (d\Psi_{II}/dx)_{x=-a/2}$, $\alpha Ee^{-\alpha a/2} = Ak\sin(ka/2)$ ... (16)

From $(d\Psi_{II}/dx)_{x=a/2} = (d\Psi_{III}/dx)_{x=a/2}$, $Ak\sin(ka/2) = \alpha De^{-\alpha a/2}$ … (17)

Dividing equation (17) by equation (16) we again get E = D. However, dividing equation (17) by equation (14),

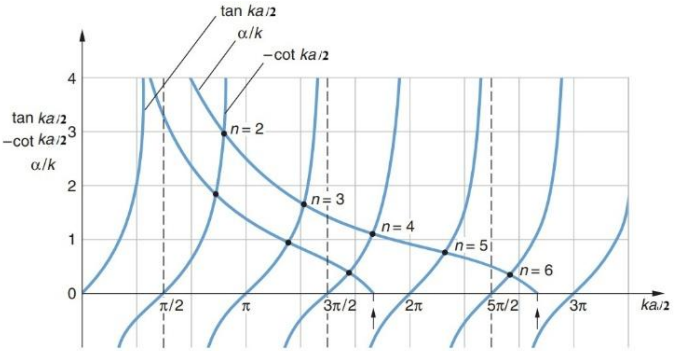
we get: $\alpha = k \tan(ka/2)$ … (18)

Since k and α depend on energy E, (18) is a transcendental equation. Similarly, picking the sine solution in region II means A = 0 and E = -D.

The corresponding transcendental equation is :

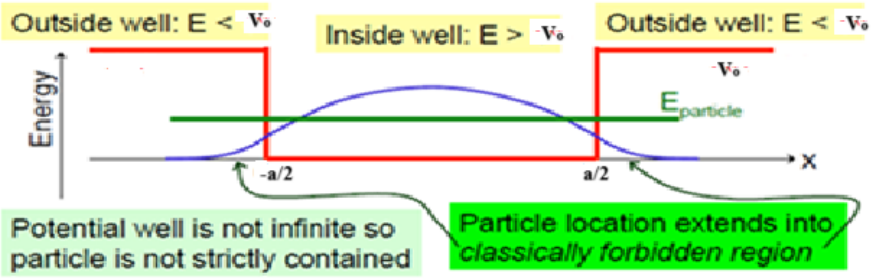
$$\alpha = -k \cot(ka/2) \quad (19)$$

The transcendental equations cannot be solved analytically, but by the graphical method of intersection of two different curves.



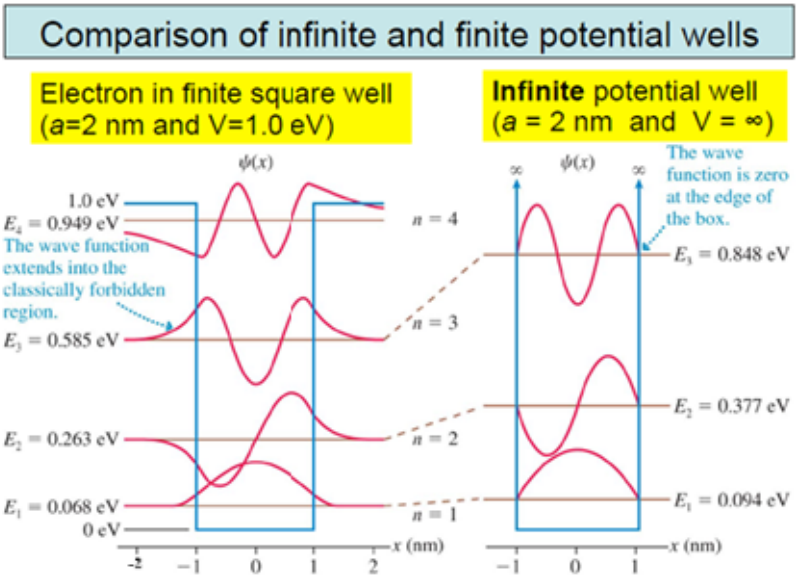
Alternatively, the two transcendental equations can be solved numerically by iterative process. The equations cannot be satisfied for any arbitrary value of E, but only a discrete set of values. The corresponding eigen function are the bound state solutions of the problem. By contrast, the energy levels E that are possible if $|E| > V_o$ is continuous.

The particles within it are not strictly confined but can now extend into the classically forbidden region I and III.



As the probability of finding the particle outside the well is finite, there is Quantum Tunneling of the particles from within the well to outside the well.

The energies of the particle in infinite well are higher than the corresponding energy levels in finite well as shown in the figure below. (This can be roughly understood by seeing that the finite well has an effective well-width higher than the infinite well width (a), hence has lower energies as per the formula $E_n = n^2h^2/8ma^2$



The concept of Quantum Tunneling as understood from the Square Well Potential helps us to understand the fields of Radioactive decay, Scanning Tunneling Microscopy which is used to study surfaces and the binding of molecules.

Optical effects of quantum confinement:

Quantum confinement occurs when the dimensions of a material (nanoparticles, quantum dots, nanowires, thin films) become comparable to or smaller than the **exciton Bohr radius**.

In bulk semiconductors: electrons and holes move almost freely; the energy bandgap Eg is fixed.

In nanostructures: confinement of carriers increases their energy → **effective bandgap increases**.

Result: Absorption and emission spectra shift to **shorter wavelengths** (higher energy photons). This is the **blue shift**.

$E_n = n^2h^2/8ma^2$, where a is the width of the well

a = confinement size,
m = effective mass of carrier,
smaller L → higher E_n. This means that in quantum dots, the electron and hole confinement increase the energy required for excitation.

Effective Mass Approximation :

The bandgap of a quantum dot of radius R can be written as:

$E(R) = E_g^{bulk} + \hbar^2\pi^2/2R^2(1/m_e^* + 1/m_h^*) - 1.8e^2/(4\pi\epsilon R)$

E_g^{bulk} : Bulk bandgap

m_e^*/ m_h^* effective masses of electron/hole

2nd term: quantum confinement energy (increases as R decreases)

3rd term: Coulomb attraction (slightly decreases energy)

Net effect is that as R decreases, E(R) > Egbulk. Thus, emission shifts to higher energy (blue shift).

Experimental Evidence of Blue shift :

Large sized (bulk) CdSe quantum dots have a band gap of 1.7 eV (red). However, as the size decreases to ~2–3 nm, emission shifts

into green/blue region. We get Tunable photoluminescence, whereby nanoparticle size determines color.

A. **Absorption Spectrum :** As the dimensionality decreases (e.g., from 3D bulk material to 2D quantum wells or 1D quantum wires and 0D quantum dots), the absorption spectrum shifts towards higher photon energies (blue shift). This is due to the quantum confinement effect, which restricts the movement of charge carriers and leads to discrete energy levels and a larger band gap. The density of states (DOS) changes drastically.

B. **Fluorescence :** It is the process where a substance absorbs electromagnetic radiation (like light) and then emits it as light of a different, usually longer, wavelength. Essentially, a material absorbs a photon of higher energy and emits a photon of lower energy, resulting in a visible glow. This happens almost instantaneously. The various stages involved are :

1. Absorption: A molecule absorbs a photon of light, causing an electron to jump to a higher energy level (excited state).
2. Excitation: The molecule is now in an unstable, excited state and quickly loses some energy through vibrations and collisions with other molecules, moving to a lower vibrational level within the excited state.
3. Emission: The electron then transitions back down to its original, lower energy level (ground state). This transition releases a photon of light, which is the fluorescence.
4. Stokes Shift: Because some energy is lost during the process, the emitted photon has a lower energy and thus a longer wavelength than the absorbed photon. This phenomenon is known as the Stokes shift.

In bulk materials, electrons can occupy a range of energy levels within a continuous band. However, when a material is reduced to the nanoscale (e.g., a quantum dot), its size becomes comparable to the exciton Bohr radius (the average distance between an electron and a hole). This confinement forces electrons to occupy only certain discrete energy levels, leading to changes in their optical properties. Thus quantum confinement has a direct effect on Fluorescence properties.

Because the energy levels in a quantum-confined structure are size-dependent, the wavelength of the emitted light (i.e., fluorescence) also changes with the size of the structure. Smaller quantum dots will have larger energy gaps and emit higher energy (shorter wavelength) light, while larger quantum dots will have smaller energy gaps and emit lower energy (longer wavelength) light.

This size-tunable fluorescence makes quantum dots excellent candidates for various applications. A few applications are:

Bioimaging: Quantum dots can be used as fluorescent tags to visualize and track biological molecules and cells.

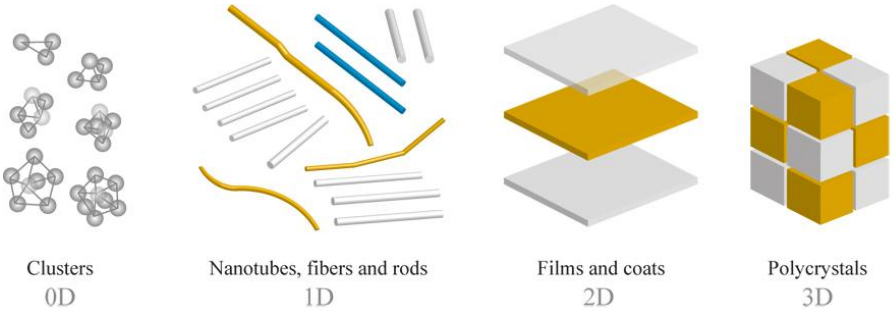
LEDs and displays: Their tunable emission colors make them useful in lighting and display technologies.

Photocatalysis: Quantum confinement can enhance the photocatalytic activity of materials, making them useful in environmental applications like water purification.

In essence, quantum confinement allows for precise control over the optical properties of materials at the nanoscale, enabling unique fluorescence characteristics and a wide range of applications.

Classification of Nanomaterials on Dimension:

Nanostructure materials are classified as zero-dimensional, one dimensional, two dimensional, three dimensional nanostructures.



(a) Zero dimensional nanomaterials

Materials wherein all the dimensions are measured within the nanoscale (no dimensions, or 0-D, are larger than 100 nm)

The best example of zero-dimensional nanomaterials is **'nanoparticles'**.

Nanoparticles can be

- Amorphous or crystalline
- Single crystalline or polycrystalline
- Composed of single or multi-chemical elements
- Exhibit various shapes and forms
- Exist individually or incorporated in a matrix
- Metallic, ceramic or polymeric

(b) Zero dimensional nanomaterials

Nanomaterials exist in one-dimensional structure. This leads to needle-like-shaped nanomaterials.

One-dimensional materials include **'nanotubes, nanorods and nanowires'**

One -dimensional nanomaterials can be

- Amorphous or crystalline
- Single crystalline or polycrystalline
- Chemically pure or impure
- Standalone materials or embedded in within another medium metallic, ceramic or polymeric.

(c) Two-dimensional nanomaterials

Two-dimensional nanomaterials exhibit plate-like shapes. Two-dimensional nanomaterials include **'nanofilms, nanolayers and nanocoatings'**.

Two-dimensional nanomaterials can be

- Amorphous or crystalline
- Made up of various chemical compositions
- Used as a single layer or as multilayer structures
- Deposited on a substrate

- Integrated in a surrounding matrix material metallic, ceramic or polymeric

(d) Three-dimensional nanomaterials

'Bulk nanomaterials' are materials that are not confined to the nanoscale in any dimension. These nanomaterials are thus characterized by having three arbitrarily dimensions above 100 nm. Materials possess a nanocrystalline structure or involve the presence of features at the nanoscale.

- In terms of nanocrystalline structure, bulk nanomaterials can be composed of a multiple arrangement of nanosize crystals, most typically in different orientations.
- In a nanoscale, three dimensional nanomaterials can contain dispersions of nanoparticles, bundles of nanowires and nanotubes as well as multinanolayers.

Nanomaterials characteristics

Category of nanomaterials	example
One-dimensional nanomaterials	layers, multi-layers, thin films, platelets and surface coatings. They have been developed and used for decades, particularly in the electronics industry.
Two-dimensional nanomaterials	nanowires, nanofibres made from a variety of elements other than carbon, nanotubes and, a subset of this group, carbon nanotubes.
Three-dimensional nanomaterials	are known as nanoparticles and include precipitates, colloids and quantum dots (tiny particles of semiconductor materials), and Nanocrystalline materials

Quantum effects

- The overall behavior of bulk crystalline materials changes when the dimensions are reduced to the nanoscale.
- For 0-D nanomaterials, where all the dimensions are at the nanoscale, an electron is confined in 3D space. No electron delocalization (freedom to move) occurs.
- Fr 1D nanomaterials, electron confinement occurs in 2D whereas delocalization takes place along the long axis of then nanowirenanorod/nanotube
- In the case of 2D nanomaterials, the conduction electrons will be confined across the thickness but delocalized in the plane of the sheet.

Electrons confinement

- For 0D nanomaterials the electrons are fully confined.
- For 3D nanomaterials the electrons are fully delocalized
- In 1D and 2D nanomaterials, electron confinement and delocalization coexist.
- The effect of confinement on the resulting energy states can be calculated by quantum mechanics as the particle in the box problem. A electron is considered to exist inside of an infinitely deep potential well from which it cannot escape and is confined by the dimensions of the nanostructure.

Quantum Structures:

When we reduce 1-dimension of a bulk material which is in 3-dimension, it results in a structure in 2-dimensions and is called **quantum film**. If the reduction takes place in 2-directions, the resultant nanomaterial is in 1-dimension and is called **quantum wire**. If the reduction takes place in all the 3-directions, the material reduces to a point called as **quantum dot**. It is also called nanoparticle or cluster.

Charge carriers are able to move in all directions in a 3-D material, confined to a plane in a film in only one direction in a quantum wire but in a 0-D structure they will remain confined to a very small space.

3-Dimension **2-Dimension** **1-Dimension**
0-Dimension **Bulk material** **Quantum film**
Quantum wire **Quantum dot**

Density of States (DOS):

When bulk materials are changed from nanomaterials to other dimensions density of states (DOS) for various quantum structures as a function of energy is as shown.

a) **DOS for Bulk material (3D)**

The variation of DOS for bulk material is parabolic and varies continuously and density is 10^{26} /m^3 eV

$$g(E)dE = \left[\frac{8\sqrt{2}\pi m^{\frac{3}{2}}}{h^3}\right] E^{\frac{1}{2}} dE$$

b) **DOS for 2-D structures**

For a 2-D structure the DOS varies as step function. DOS varies at E1, E2, E3 etc. It is due to the fact that first sub band continues even when second sub band starts etc. Hence DOS rises at once at the beginning

of second sub band. The locus of all the corners of the steps will be a parabola. If the thickness of the 2-D material is increased more sub-bands will be formed which results in decrease of the step width. On further increase in thickness step width decreases and finally DOS becomes parabolic.

The density of States for 2-D structure is of the order $10^{18}/m^2$ eV.

$$D(E)\,dE = \frac{m^*}{\pi\hbar^2}\sum H(E-E_i)\,dE$$

Where m* is the effective mass of electron H(E-Ei) is a step function called Heaviside function, its values are zero for E< Ei and 1 for E greater than Ei.

c) **DOS 1-D Structure**

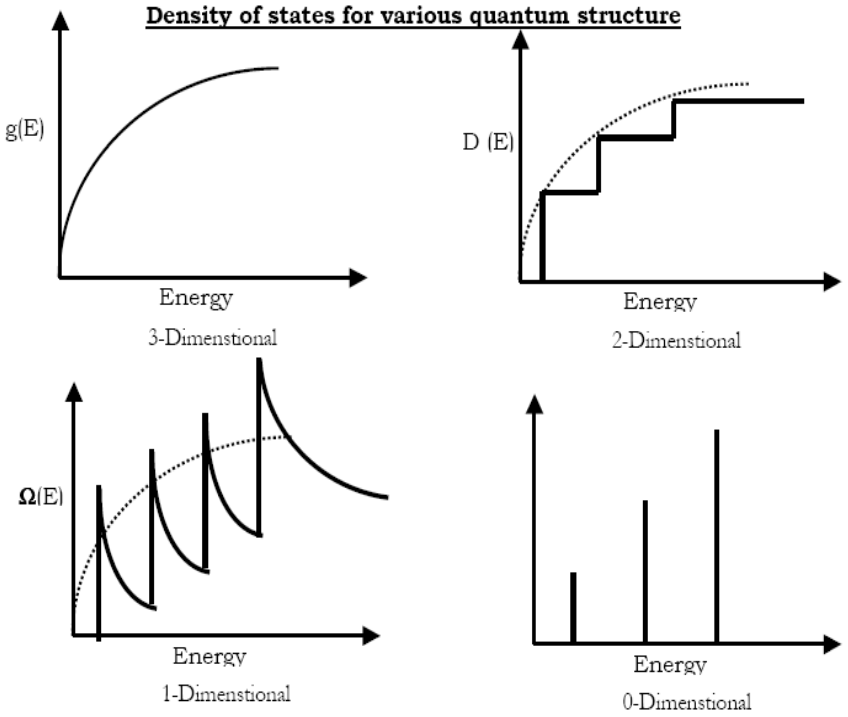
For a quantum wire of 1-D structure, the DOS variation is smooth there are density peaks at different energy values and decreases rapidly in between.

$$D(E)\,dE = \frac{\sqrt{2m}}{\pi\hbar^2}\sum \frac{n_i\,H(E-E_i)}{\sqrt{(E-E_i}}\,dE$$

Where H(E-Ei) is Heaviside function, n_i is degeneracy factor. The density of the order of $10^9/m^2$ eV

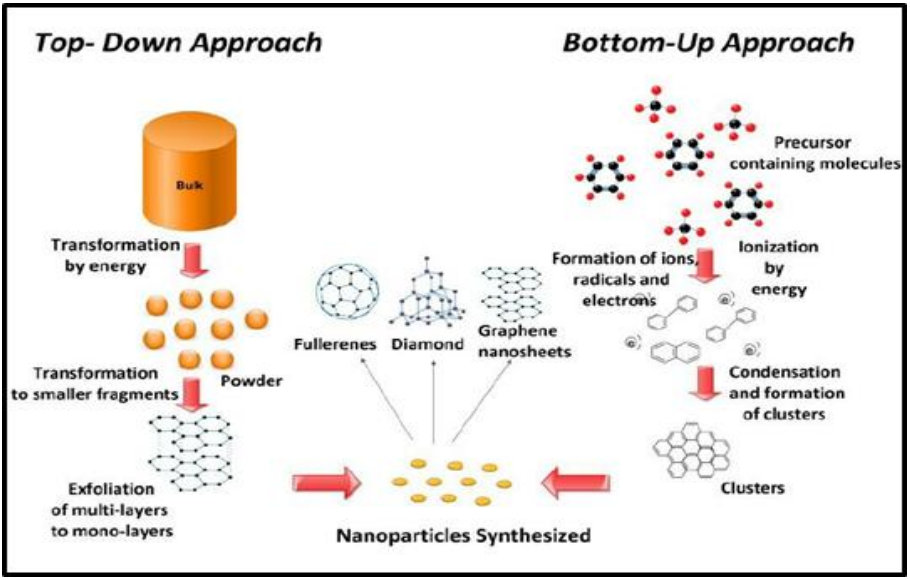
d) **DOS for 0-D structures**

For a 0-D structure, the permitted energy values are just discrete because of confined condition for the electrons.



Synthesis of Nanomaterials

During the last decade, various methods were developed for the synthesis of nanostructure materials in different forms like colloids, clusters, powders, tubes wires, thin films etc. The different synthesis methods are Ball milling, sol-gel, inverse micelles, chemical vapour deposition, micro emulsion, hydrothermal, combustion, Co-precipitation, laser ablation, sputtering, electric arc deposition; etc. There are two approaches, followed in preparing nanomaterials. One is known as **top-down approach** and the other is called **bottom-up approach**



A) Bottom-up approach:

In bottom-up approach nanoparticles are prepared by bringing individual **atoms and molecules together** until nano size is reached. Bottom-up approach refers to the build up of a material from the bottom: atom-by-atom, molecule-by-molecule, or cluster-by cluster. Many of these techniques are still under development or are just beginning to be used for commercial production of nano powders. Example: Organometallic chemical route, revere-micelle route, sol-gel synthesis, colloidal, hydrothermal synthesis, template-assisted sol-gel, electrodeposition, etc

Advantages

- Ultra-fine nanoparticles, nanoshells, nanotubes can be prepared

- Deposition parameters can be controlled
- Size and shape of nanomaterials can be controlled
- Easier process
- Narrow size distribution is possible (1-20 nm) • Cheaper technique

Disadvantages

- Large scale production is difficult
- Chemical purification of nanoparticles is required

a) Sol Gel method

The sol-gel process, involves the evolution of inorganic networks through the formation of a colloidal suspension (sol) and gelation of the sol to form a network in a continuous liquid phase (gel). The sol-gel method of synthesizing nanomaterial is very popular amongst chemists and is widely employed to prepare oxide materials.

The sol-gel process can be characterized by a series of distinct steps.

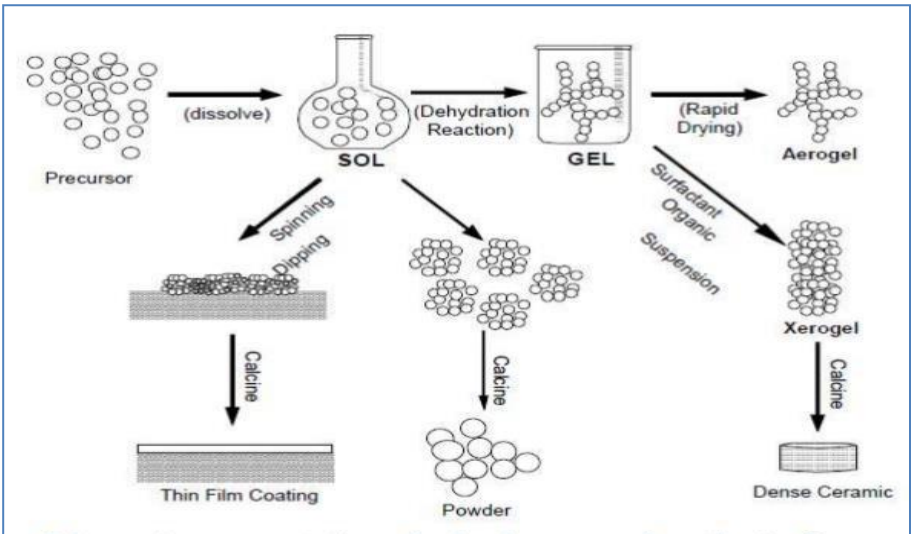
Step 1: Formation of different stable solutions of the alkoxide or solvated metal precursor (**the sol**).

Step 2: Gelation resulting from the formation of an oxide- or alcohol-bridged network (the gel) by a polycondensation or polyesterification reaction that results in a dramatic increase in the viscosity of the solution.

$MOR + H_2O \rightarrow MOH + ROH$ (hydrolysis)

$MOH + ROM \rightarrow M\text{-}O\text{-}M + ROH$ (condensation)

Step 3: Aging of the gel (**Syneresis**), during which the polycondensation reactions continue until the gel transforms into a solid mass, accompanied by contraction of the gel network and expulsion of solvent from gel pores. The aging process of gels can exceed 7 days and is critical to the prevention of cracks in gels that have been cast.



Schematic representation of sol-gel process of synthesis of nanomaterials

Step 4: Drying of the gel, when water and other volatile liquids are removed from the gel network. This process is complicated due to fundamental changes in the structure of the gel. The drying process has itself been broken into four distinct steps:

(i) the constant rate period
(ii) the critical point
(iii) the falling rate period
(iv) the second falling rate period.

If isolated by thermal evaporation, the resulting monolith is termed a *xerogel*. If the solvent (such as water) is extracted under supercritical or near supercritical conditions, the product is an *aerogel*.

Step 5: Dehydration, during which surface- bound M-OH groups are removed, thereby stabilizing the gel against rehydration. This is normally achieved by calcining the monolith at temperatures up to 800 °C.

Step 6: Densification and decomposition of the gels at high temperatures (T > 800 °C). The pores of the gel network are collapsed, and remaining organic species are volatilized. The typical steps that are involved in sol-gel processing are shown in the schematic diagram below.

Advantages:

1. The main advantage is one can get monosized nano particles.
2. Can produce thin bond-coating to provide excellent adhesion between the metallic substrate and the top coat.
3. Can produce thick coating to provide corrosion protection performance.
4. Can easily shape materials into complex geometries in a gel state.
5. Can produce high purity products
6. Can have low temperature sintering capability, usually 200-600°C.
7. Can provide a simple, economic and effective method to produce high quality coatings.

Disadvantages:

1. Controlling the growth of the particles and then stopping the newly formed particles from agglomerating.
2. Other technical issues are ensuring the reactions are complete so that no unwanted reactant is left on the product and completely removing any growth aids that may have been used in the process.
3. Production rates of nano powders are very low by this process.

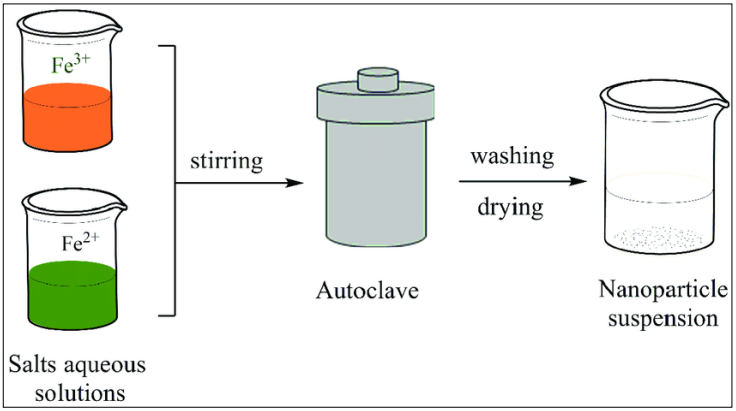
Applications:

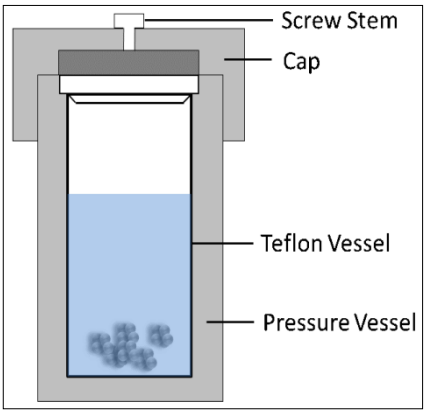
1. It can be used in ceramics manufacturing processes, as an investment casting material, or as a means of producing very thin films of metal oxides for various purposes.
2. Sol-gel derived materials have diverse applications in optics, electronics, energy, space, (bio) sensors, medicine (e.g. controlled drug release) and separation (e.g. chromatography) technology. One of the more important applications of sol-gel processing is to carry out zeolite synthesis.
3. Other elements (metals, metal oxides) can be easily incorporated into the final product and the silicalite sol formed by this method is very stable.
4. Other products fabricated with this process include various ceramic membranes for microfiltration, ultrafiltration, nanofiltration, pervaporation and reverse osmosis.

b) <u>Hydrothermal synthesis method</u>

hydrothermal processing is "*any heterogeneous chemical reaction in the presence of a solvent (aqueous or non-aqueous) above room temperature and at pressures > 1 atm in a closed system*". Under hydrothermal conditions a broad range of nanostructure materials might be formed. Hydrothermal methods rely on the ability of water to hydrolyze and dehydrate metal salts on elevated temperature.

- The hydrothermal reactions are performed in aqueous media in reactors or autoclaves where the pressure is higher than 1.4×10^4 kPa at temperatures above 200 °C.
- The solubility of the resulting metal oxides in water may be very low to produce the supersaturation.





Step 1: prepare aqueous solution by dissolving the precursor in the solvent (water).

Step 2: Transform the solution to the autoclaves and kept it in hot air oven During hydrothermal this time metal cations initially precipitate in the form of polymeric hydroxides. Over time, these hydroxides undergo dehydration to form the metal oxide crystal structures. The time has a more significant influence on the average particle size than concentration; shorter time favours monodispersed particles.

Step 3: allow autoclaves to cool down to normal room temperature

Step 4: purification procedure has to carried out using filtration, dialysis, centrifugation etc….finally dry the obtained powder

The size and nature of the prepared nanoparticles strongly depends on the pH, temperature, retention time, concentration of the complexing agent (Ex: oxalic acid), cationic surfactant, and alkali sources.

Advantages

- simple, economic process
- good crystallinity of prepared nanomaterials.
- Uniform particles will be formed
- It is a preferred method for making nanoceramic materials economically.
- Hydrothermal synthesis can generate nanomaterials which are not stable at elevated temperatures
- Nanomaterials with high vapor pressures can be produced by the hydrothermal method with minimum loss of materials
- relatively mild operating conditions
- impurities will be very less

Disadvantages

- difficult to produce large quantity of nanomaterials
- large scale production of nanomaterials is very difficult

Applications

- prepare oxide materials for electronic devices
- dielectric thin film industry
- ceramic industry
- semiconductor industry
- electronic industry
- aerospace engineering

B) Top down approach:

In the top-down approach, the material in bulk form is **broken down** to smaller and smaller units until the nano size is obtained**.**

Top-down approach involves the breaking down of the bulk material into nanosized structures or particles. Top-down synthesis techniques are extension of those that have been used for producing micron sized particles. Top-down approaches are inherently simpler and depend either on removal or division of bulk material or on miniaturization of bulk fabrication processes to produce the desired structure with appropriate properties. The biggest problem with the top-down approach is the imperfection of surface structure. For example, nanowires made by lithography are not smooth and may contain a lot of impurities and structural defects on its surface.

Examples of such techniques are

Example: ***Ball milling, Lithography*** and **Laser ablation.**

- high-energy
- wet ball milling,
- electron beam lithography,
- atomic force manipulation,
- gas-phase condensation,
- aerosol spray

Advantages

- Dominant in micro fabrication, well-developed techniques
- Low cost and cheaper
- Used in industries
- Easy processing
- producing nanostructures with less defects and more homogenous chemical composition
- the cost and controlled shape and size of the product

Disadvantages

- imperfection of surface structure
- non-uniform particles sizes
- difficulty to make uniform particle size
- There are less tools for handling atoms and molecules.
- Needs compatible surfaces and molecules

1) **Ball Milling Method (Top Down approach)**

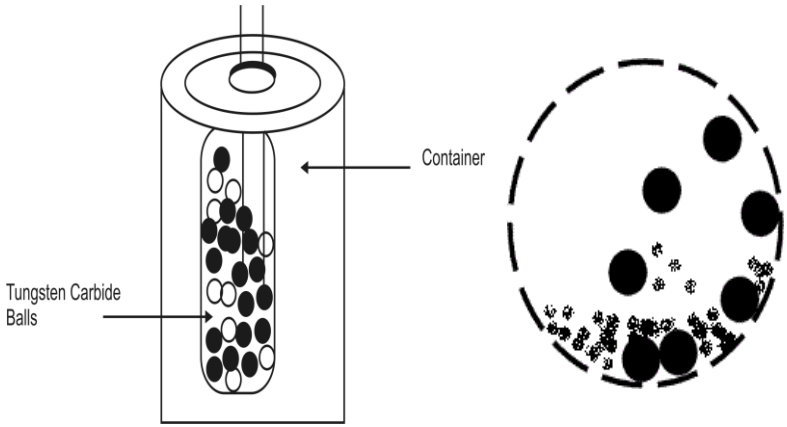
There are many types of mills namely, Planetary, vibratory, tumbler etc. depending on the requirements. In this method, a powder mixture placed in the ball mill is subjected to high-energy collision from the balls. This process was developed by **Benjamin and his co-workers** at the International Nickel Company in the late of 1960.

Principle:

Small hard balls are allowed to rotate inside a container and then it is made to fall on a solid with high force to crush the solid into nanocrystal.

Construction and Working:

Hardened steel or tungsten carbide balls are put in a container along with powder of particles (50mm) of a desired material. The container is closed with tight lids. When the container is rotating around the central axis; the material is forced to press against the walls. The milling balls impart energy on collision and produce smaller grain size of nano particle. Ball milling is also known as mechanical alloying or crushing.



Advantages of ball milling

1. Nanopowders of 2 to 20 nm in size can be produced. The size of nanopowder also depends upon the speed and duration of rotation of the balls.
2. It is an inexpensive and easy process.
3. Suitable for large scale production.

Disadvantages of ball milling

1. As the process is not so sophisticated, therefore the shape of the nano material is irregular this will adversely affect properties of the resulting product.
2. There may be contaminants inserted from balls and milling additives.
3. This method produces crystal defects.

Applications:

1. Ball milling method is useful in preparation of elemental and metal oxide nano crystals like Co, Cr, Al-Fe, Ag-Fe and Fe.
2. Variety of intermetallic compounds of Ni and Al can be formed.

3. Ball milling method is useful in producing new type building materials, fire – proof materials, glass ceramics, etc.

2) Lithography

It is a process of **pattern transfer** onto a material's surface at micro- and nanoscale. It is a **top-down approach**: starting with bulk material (substrate) and *removing or modifying portions* to create nanoscale structures and is widely used in **semiconductor industry**, **nanodevices**, **sensors**, and **nanoelectronics**.

Basic Steps involved in Lithography are :

1. **Substrate preparation** – usually silicon wafer or other suitable base.
2. **Coating with resist** – apply a thin film of photoresist/electron resist.
3. **Exposure** – illuminate with radiation (UV, electron beam, ion beam, X-rays) through a mask or direct writing.
4. **Development** – exposed regions are dissolved (positive resist) or unexposed regions are dissolved (negative resist).
5. **Etching/Deposition** – material under patterned resist is etched away or filled with another material.
6. **Lift-off** – resist is removed, leaving behind nanostructures.

Types of Lithography are :

a) Photolithography – UV light + mask; standard for microelectronics, limited to ~100 nm due to diffraction.

b) Electron-Beam Lithography (EBL) – direct writing with focused electron beam; very high resolution (<10 nm), but slow.
c) Ion-Beam Lithography – uses focused ions (e.g., Ga+); precise but expensive.
d) X-Ray Lithography – shorter wavelength → higher resolution (~20 nm).
e) Nanoimprint Lithography – uses physical molds to stamp nanoscale patterns (cost-effective).

Advantages of Lithography

a) High precision in **defining patterns** (down to few nanometers).
b) Compatible with **large-scale manufacturing** (especially photolithography).
c) Enables integration into **devices and circuits**.

Limitations of Lithography

a) Equipment is **expensive and complex**.
b) **Resolution limited** by wavelength (photolithography).
c) Generally **2D patterning**; difficult for 3D nanostructures.
d) Requires **cleanroom environment**.

Applications of Lithography

a) Semiconductors & ICs (chip fabrication).
b) MEMS/NEMS devices.
c) Nanophotonic structures.
d) Sensors, waveguides, quantum devices.

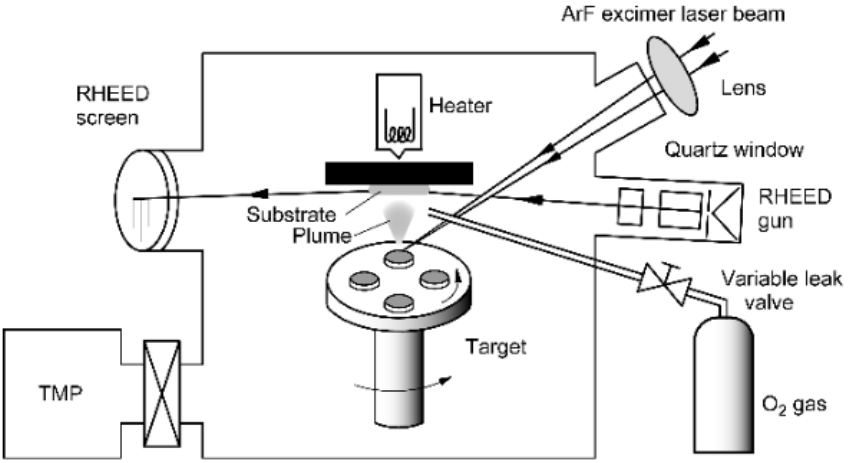
3) Laser ablation method

Pulsed Laser Deposition (PLD) is a thin film deposition technique that is used to deposit materials on substrates. A base system consists of a target,

Step1 : A high-energy focused laser beam is capable of eroding the surface of a target rotating with a velocity of one revolution per second.

Step 2: The material is vaporized on the substrate, and as a result, a film is produced on it at a rate of 0.07 nm/ laser pulse.

Step 3: The growth canbe supported by heating the substrate (750 °C) and by chemical reactions (oxygen at 50Pa).



Laser ablation system

Mechanism

The removal of material from solid surfaces by the technique of laser ablation is known as laser plume ablation (LPA). The raw material is provided as a solid. Its dissolution is achieved by a focused laser beam. Deposition takes place in a ultra-high vacuum or in the presence of

some neutral gas such as Argon at a certain pressure. The gas at the desired pressure is used to control the deposition rate as well as film composition. A high power pulsed laser is focused onto the target material and, when the laser fluence (power density) exceeds the ablation threshold value for the material, chemical bonds are broken and the material is fractured into energetic fragments, typically a mixture of neutral atoms, molecules, and ions. Since these fragments are highly energetic, the process is considered as an explosive evaporation. Most of the absorbed energy is deposited in the ejected material, so that there is little or no thermal damage to the surrounding target material. The vapor (plume) is collected on a substrate placed a short distance from the target.

Advantages

- Ultrapure films or particles can be produced by Laser ablation since it uses a vacuum environment.
- Laser ablation can provide good structural control by careful monitoring of the processing conditions.
- Materials can be deposited with improved properties compared to the substrate material.
- Almost any type of inorganic material can be used as well as some kinds of organic materials.
- The process is more environmentally friendly than processes such as electroplating.

Disadvantages

- Since Laser ablation operates in a low pressure range, it increases the complexity of deposition and cost of production.
- It is a line of sight technique meaning that it is extremely difficult to coat undercuts and similar surface features.
- High capital cost.

- Some processes operate at high vacuums and temperatures requiring skilled operators.
- Processes requiring large amounts of heat require appropriate cooling systems.
- The rate of coating deposition is usually quite slow.

Nanoscale Semiconductors and Quantum Dots

Nanoscale Semiconductors : When **semiconductor crystals** (e.g., Si, GaAs, CdSe) are reduced to sizes of **1–100 nm**, they exhibit **quantum confinement effects**. At this scale, the **electron wavefunction** is confined, and classical band theory is modified.
Density of States (DOS) changes as follows:

- Bulk (3D): continuous bands.
- Thin films (2D): step-like DOS.
- Nanowires (1D): sharp peaks.
- Quantum Dots (0D): discrete atomic-like levels.

Quantum Dots (QDs)
Semiconductor nanocrystals (2–10 nm) where electrons and holes are confined in all 3 dimensions, are called Quantum dots or "**artificial atoms**" due to discrete energy levels.

- Example materials: CdSe, PbS, InAs, ZnS.
- **Synthesis**: colloidal methods, epitaxial growth, lithography.

Due to Quantum confinement effects, these nanoscale semiconductors and Quantum Dots exhibit size-dependent Optical and Electrical Properties. As dot size decreases, the band gap increases. This means that the **Emission/absorption wavelength** shifts.

a) Larger QD → smaller band gap → red-shift emission.

b) Smaller QD → larger band gap → blue-shift emission.

This enables **tunable fluorescence** across visible spectrum.

- Applications: quantum dot displays (QLED TVs), bio-imaging markers, lasers.

The important electrical properties that are observed due to quantum confinement are

- Discrete electronic states → **Coulomb blockade effect** (charge quantization in QDs).
- Carrier mobility decreases at nanoscale due to **surface scattering**.
- **Tunneling effects** dominate transport (electrons can tunnel between QDs).
- **Single Electron Transistors (SETs)** exploit charge quantization in QDs.

These quantum confined structures find wide applications in

- **Optoelectronics**: QD-LEDs, solar cells, lasers.
- **Biomedical applications**: fluorescent tags for imaging.
- **Quantum Computing**: QD spin qubits.
- **Sensing**: photodetectors, chemical sensors.

Q.No	Sample Questions	CO
1.	State De Broglie hypothesis.	1
2.	What is wave function? Give its physical significance and properties.	1
3.	State Heisenberg's uncertainty principle. By applying Heisenberg's uncertainty principle, illustrate the broadening of spectral lines.	1&2
4.	What are Eigen functions and Eigen values?	1

5.	Setup time independent one-dimensional Schrodinger's wave equation for a matter wave.	**1**
6.	Apply the time independent Schrodinger's wave equation to find the solutions for a particle in an infinite potential well of width 'a'. Hence obtain normalized wave function.	**1&2**
7	Solve the Schrodinger's wave equation for a free particle.	**1&2**
8	Set up the differential equation for a particle in a 1D well of finite depth and arrive at an Eigen Value expression.	**2**

PNo.	Problems	CO
1.	Calculate the de Broglie wavelength associated with a proton moving with a velocity equal to $(1/20)^{th}$ of the velocity of light. **To be found**: de Broglie wavelength, λ **Solution:** $\lambda = \frac{h}{mv} = \frac{6.625\times10^{-34}}{1.67\times10^{-27}\times(1/20)\times3\times10^{8}} = 2.64\times10^{-14}\,m$	3
2.	An electron and a proton are accelerated through the same potential difference. Find the ratio of their de Broglie wavelengths. **To be found:** Ratio of de Broglie wavelength, λ **Solution:** **De Broglie Wavelength, λ** $\lambda = \frac{h}{\sqrt{2mE}}$, $\therefore \lambda \,\alpha\, \frac{1}{\sqrt{m}}$ **For electron,** $\lambda_e \,\alpha\, \frac{1}{\sqrt{m_e}}$ **For proton,** $\lambda_p \,\alpha\, \frac{1}{\sqrt{m_p}}$ **Ratio of De Broglie Wavelengths,** $\therefore \frac{\lambda_e}{\lambda_p} = \frac{\sqrt{m_p}}{\sqrt{m_e}}$	3
3.	Compare the energy of a photon with that of a neutron when both are associated with wavelength of $1A^o$. Given that the mass of neutron is 1.67×10^{-27}kg.	3

	To be found: Comparison of energy of photon with that of neutron **Solution:** **Energy of neutron**, $E_n = \frac{P^2}{2m} = \frac{h^2}{2\lambda^2 m} = \frac{(6.625\times10^{-34})^2}{2\times(1\times10^{-10})^2\times1.67\times10^{-27}} = 13.1\times10^{-21} J$ =0.082eV **Energy of photon**, $E_p = \frac{hc}{\lambda} = \frac{6.625\times10^{-34}\times3\times10^{8}}{1\times10^{-10}} = 19.89\times10^{-16} J$ =12421.9eV **Ratio of energies**, $\frac{E_p}{E_n} = 1.5148\times10^{5}$	
4.	An electron has a speed of 4.8 x 10^5 m/s accurate to 0.012 %. With what accuracy with which its position can be located. **To be found**: Uncertainty in position, Δx **Solution:** **Uncertainty principle** is given by, $\Delta x \Delta p \geq \frac{h}{4\pi}$ **Uncertainty in speed**, $\Delta v = 4.8 \times 10^5 \times \frac{0.012}{100} = 57.6 m/s$ **Uncertainty in position**, $\Delta x = \frac{h}{4\pi m \Delta v} = \frac{6.625\times10^{-34}}{4\times3.14\times9.1\times10^{-31}\times57.6} = 1\times10^{-6} m$	3

5.	The inherent uncertainty in the measurement of time spent by Iridium-191 nuclei in the excited state is found to be 1.4×10^{-10} s. Estimate the uncertainty that results in its energy in the excited state. **To be found**: Uncertainty in energy, ΔE **Solution:Uncertainty principle** is given by, $\Delta E \Delta t \geq \frac{h}{4\pi}$ $\therefore \Delta E \geq \frac{h}{4\pi\Delta t} \geq \frac{6.625\times10^{-34}}{4\times3.14\times1.4\times10^{-10}} \geq 3.77\times10^{-25} J$	3
6.	The position and momentum of 1 keV electron are simultaneously determined and if its position is located within 1Å. What is the percentage of uncertainty in its momentum? **To be found**: Percentage of uncertainty in momentum of electron, Δp **Solution:** Uncertainty principle is given by, $\Delta x \Delta p \geq \frac{h}{4\pi}$ **Uncertainty in momentum**, $\Delta p = \frac{h}{4\pi\Delta x} = \frac{6.625\times10^{-34}}{4\times3.14\times1\times10^{-10}} = 0.53\times10^{-24} kg.m/s$ **Momentum**, $p = \sqrt{2mE} = \sqrt{2\times9.1\times10^{-31}\times1\times10^{-16}} = 1.707\times10^{-23} kgm/s$ **Percentage of uncertainty in momentum of electron**, $\frac{\Delta p}{p}\times100 = 3.1$	3

7.	Show that the energy Eigen value of a particle in second excited state is equal to 9 times the zero point energy. **To be found: E**nergy Eigen value for second excited state is equal to 9 times the zero point energy. **Solution: Energy Eigen value equation is given by,** $E = \frac{n^2h^2}{8ma^2}$ **n=1, zero-point energy state,** $E_1 = \frac{h^2}{8ma^2}$ **n=3, second excited state,** $E_3 = \frac{9h^2}{8ma^2} = 9E_1$	3
8.	An electron is bound in a one-dimensional potential well of width 1Å, but of infinite height. Find the energy value for the electron in the ground state. **To be found:** Energy Eigen value **Solution:** Energy Eigen value equation is given by, $E = \frac{n^2h^2}{8ma^2}$ **For n=1, ground state energy,** $E_1 = \frac{\left(6.625\times10^{-34}\right)^2}{8\times9.1\times10^{-31}\times\left(10^{-10}\right)^2} = 37.65eV$	

9.	An electron is bound in one dimensional potential well of infinite potential of width 0.12 nm. Find the energy values in the ground state and also the first two exited state. **To be found:** Energy Eigen value **Solution:** Energy Eigen value equation is given by, $E = \frac{n^2h^2}{8ma^2}$ **For n=1, ground state energy,** $E_1 = \frac{\left(6.625\times10^{-34}\right)^2}{8\times9.1\times10^{-31}\times\left(0.12\times10^{-9}\right)^2} = 26.16eV$ **For n=2 and 3,first and second excited state,** $E_2 = 4E_1 = 104.61eV \;\; E_3 = 9E_1 = 235.44eV$	
10.	An electron is trapped in a potential well of a width 0.5nm. If a transition takes place from the first excited state to the ground state find the wavelength of the photon emitted. **To be found**: Wavelength of the photon emitted, λ **Solution:** For n=1, ground state energy, $E_1 = \frac{\left(6.625\times10^{-34}\right)^2}{8\times9.1\times10^{-31}\times\left(0.5\times10^{-9}\right)^2} = 2.41\times10^{-19}J = 1.507eV$ For n=2,first excited state,	

	$E_2 = 4E_1 = 9.64\times10^{-19} J = 6.025 eV$ Energy difference, $\Delta E = E_2 - E_1 = 7.23\times10^{-19} J = 4.518 eV$ **Wavelength of the photon emitted** , $\lambda = \frac{hc}{\Delta E} = \frac{6.625\times10^{-34}\times3\times10^{8}}{7.23\times10^{-19} J} = 274.8 nm$	